

CompAstr26 – Coding Exercise

Particle sampling of a cored galaxy density profile and mass assignment on a grid

Density profile, parameters and units

The galaxy mass distribution has density profile

$$\rho(r) \approx \frac{\rho_0}{1 + (r/r_0)^2}, \quad (1)$$

where

- ρ_0 is the finite central core density;
- r_0 is the core scale radius, defined as

$$r_0 = \sqrt{\frac{9 \sigma_v^2}{4\pi G \rho_0}}. \quad (2)$$

Assume:

- a core radius $r_0 = 1$ kpc;
- a velocity scale $\sigma_v = 150$ km/s;
- a maximum radius of 250 kpc.

Use a system of units in which velocities are measured in km/s and radius is in kpc.

Tasks

1. Particle distribution. Construct a numerical model of a galaxy mass distribution with the density profile given by Eq. (1). Distribute the particles using a random sampling of the given density profile, so that one tenth of the core radius and the maximum radius are the minimum and maximum distance of the particles from the origin of a reference coordinate system (use this to distribute the particles in space). Assume particles have all the same mass.

2. Grids. Construct Cartesian grids (3D) of

- 100^3 cells (type A);
- 200^3 cells (type B).

Choose the size of the grid appropriately in coordinate space, based on the size of the system and the units provided.

3. Mass assignment. Map the particle distribution on the grid using

- Nearest Grid Point (NGP);
- Cloud In Cell (CIC).

4. Comparison. Compare how the two methods sample better or worse the original density profile.